

```
apiVersion: v1
kind: Pod
metadata:
  name: test-vol
spec:
  containers:
  - image: webserver
    name: web-container
    volumeMounts:
    - mountPath: /cache
      name: cache-volume
  volumes:
  - name: cache-volume
    emptyDir:
      sizeLimit: 250Mi
```

```
apiVersion: v1
kind: Pod
metadata:
  name: cm-pod
spec:
  containers:
  - name: test
    image: busybox:1.28
    command: ['sh', '-c', 'echo "The app is running!" && tail -f /dev/null']
    volumeMounts:
    - name: config-vol
      mountPath: /etc/config
  volumes:
  - name: config-vol
    configMap:
      name: log-config
      items:
      - key: log_level
        path: log_level.conf
```

```
apiVersion: v1
kind: Pod
```

```
metadata:
  name: downwardapi-pod
  labels:
    app: demo-app
spec:
  containers:
  - name: demo-container
    image: busybox
    command: ["sh", "-c", "echo POD_NAME=$POD_NAME; echo
POD_NAMESPACE=$POD_NAMESPACE; cat /etc/pod-info/labels; cat /etc/pod-
info/cpu_limit; sleep 3600"]
```

```
  env:
  - name: POD_NAME
    valueFrom:
      fieldRef:
        fieldPath: metadata.name
  - name: POD_NAMESPACE
    valueFrom:
      fieldRef:
        fieldPath: metadata.namespace
```

```
  volumeMounts:
  - name: pod-info
    mountPath: /etc/pod-info
```

```
  volumes:
  - name: pod-info
    downwardAPI:
      items:
      - path: "labels"
        fieldRef:
          fieldPath: metadata.labels
      - path: "cpu_limit"
        resourceFieldRef:
          containerName: demo-container
          resource: limits.cpu
```

```
apiVersion: v1
kind: Pod
metadata:
```

```
  name: projected-volume
spec:
  containers:
  - name: container-test
    image: busybox:1.28
    command: ["sleep", "3600"]
    volumeMounts:
    - name: all-in-one
      mountPath: "/projected-volume"
      readOnly: true
  volumes:
  - name: all-in-one
    projected:
      sources:
      - secret:
          name: secret-user
          items:
          - key: username
            path: group/username
      - secret:
          name: secret-pwd
          items:
          - key: password
            path: group/password
```

```
apiVersion: v1
kind: Pod
metadata:
  name: hostpath-pod
spec:
  containers:
  - name: web-container
    image: registry.k8s.io/webserver
    volumeMounts:
    - mountPath: /foo
      name: example-volume
      readOnly: true
  volumes:
  - name: example-volume
    hostPath:
      path: /data/foo
      type: Directory
```

Example Pod definition with PersistentVolumeClaim volume:

```
apiVersion: v1
kind: Pod
metadata:
  name: pv-pod
spec:
  volumes:
    - name: pv-storage
      persistentVolumeClaim:
        claimName: pv-claim
  containers:
    - name: pv-container
      image: nginx
      ports:
        - containerPort: 80
          name: "http-server"
      volumeMounts:
        - mountPath: "/usr/share/nginx/html"
          name: pv-storage
```

```
apiVersion: v1
kind: Pod
metadata:
  name: nfs-pod
spec:
  containers:
    - image: registry.k8s.io/webserver
      name: web-container
      volumeMounts:
        - mountPath: /nfs-data
          name: nfs-volume
  volumes:
    - name: nfs-volume
      nfs:
        server: nfs-server.example.com
        path: /nfs-vol
        readOnly: true
```

apiVersion: snapshot.storage.k8s.io/v1
kind: VolumeSnapshot
metadata:
 name: snapshot-test
spec:
 volumeSnapshotClassName: csi-hostpath-snapclass
 source:
 persistentVolumeClaimName: pvc-test

apiVersion: v1
kind: PersistentVolumeClaim
metadata:
 name: restored-pvc
spec:
 dataSource:
 name: db-snapshot
 kind: VolumeSnapshot
 apiGroup: snapshot.storage.k8s.io
 storageClassName: csi-storage-class
 accessModes:
 - ReadWriteOnce
resources:
 requests:
 storage: 5Gi

apiVersion: v1
kind: PersistentVolume
metadata:
 name: demo-pv
spec:
 capacity:
 storage: 10Gi
 accessModes:
 - ReadWriteOnce
 persistentVolumeReclaimPolicy: Retain
 hostPath:
 path: /mnt/data

apiVersion: v1
kind: PersistentVolumeClaim
metadata:
 name: demo-pvc
spec:
 accessModes:
 - ReadWriteOnce
 resources:
 requests:
 storage: 5Gi
 storageClassName: standard

kind: PersistentVolume
spec:
 capacity:
 storage: 10Gi
 accessModes:
 - ReadWriteOnce
 storageClassName: manual

kind: PersistentVolumeClaim
spec:
 accessModes:
 - ReadWriteOnce
 resources:
 requests:
 storage: 5Gi
 storageClassName: manual

volumeBindingMode: Immediate

volumeBindingMode: WaitForFirstConsumer

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: standard
provisioner: example.com/example
parameters:
  type: pd-standard
volumeBindingMode: WaitForFirstConsumer
allowedTopologies:
- matchLabelExpressions:
  - key: topology.kubernetes.io/zone
    values:
    - us-central-1a
    - us-central-1b

resources:
  requests:
    storage: 10Gi
```

```
resources:
  limits:
    storage: 20Gi
```

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: data-volume
spec:
  accessModes:
  - ReadWriteOnce
resources:
  requests:
    storage: 10Gi
  limits:
    storage: 20Gi
storageClassName: standard
```

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: data-pvc
spec:
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 5Gi
  storageClassName: standard
```

```
kubectl apply -f pvc.yaml
```

```
kubectl get pvc
```

```
volumes:
  - name: storage
    persistentVolumeClaim:
      claimName: data-pvc
```

```
containers:
  - name: app
    image: nginx
    volumeMounts:
      - mountPath: "/usr/share/nginx/html"
        name: storage
```

```
kubectl edit pvc data-pvc
```

```
kubectl delete pvc data-pvc
```

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: sc
```

annotations:
 storageclass.kubernetes.io/is-default-class: "false"
provisioner: csi-driver.example
reclaimPolicy: Retain # default value is Delete
allowVolumeExpansion: true
mountOptions:
 - discard
volumeBindingMode: WaitForFirstConsumer
parameters:
 guaranteedReadWriteLatency: "true" # provider-specific

apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
 name: ebs-sc
provisioner: ebs.csi.aws.com
volumeBindingMode: WaitForFirstConsumer
parameters:
 csi.storage.k8s.io/fstype: xfs
 type: io1
 iopsPerGB: "50"
 encrypted: "true"
 tagSpecification_1: "key1=value1"
 tagSpecification_2: "key2=value2"
allowedTopologies:
 - matchLabelExpressions:
 - key: topology.ebs.csi.aws.com/zone
 values:
 - us-east-2c

spec:
 storageClassName: "high-iops"

spec:
 storageClassName: ""

apiVersion: storage.k8s.io/v1

```
kind: StorageClass
metadata:
  name: sc-delete
provisioner: example.com/example
reclaimPolicy: Delete
```

```
apiVersion: v1
kind: PersistentVolume
metadata:
  name: retain-pv
spec:
  capacity:
    storage: 10Gi
  accessModes:
    - ReadWriteOnce
  persistentVolumeReclaimPolicy: Retain
```

```
resourcequotas:
  requests.storage: 100Gi
```